ENME 408 Spring 2026
Project 3

Ethan Hardesty
Christopher Appiah

Honor code:
"I have neither received nor given aid to anyone outside my team during this project."

1. **Waypoint generation using a second-order Hilbert curve.**

Construct a second-order Hilbert curve in the horizontal plane and use its vertices as a sequence for the translational position reference. Denote the reference position by

$$p_{\text{ref}}(t) = \begin{bmatrix} p_{1,\text{ref}}(t) \\ p_{2,\text{ref}}(t) \\ p_{3,\text{ref}}(t) \end{bmatrix}. \tag{1}$$

Let $p_{1,\text{ref}}$ and $p_{2,\text{ref}}$ follow a second-order Hilbert curve with branch length equal to 2 m. Thus, consecutive horizontal waypoints must differ by 2 m along either the p_1 or p_2 direction. Let the vertical component vary during the maneuver so that:

- during the first half of the trajectory, $p_{3,\text{ref}}$ increases by by 1 m, and
- item during the second half of the trajectory, $p_{3,\text{ref}}$ decreases by 1 m.

Furthermore, consider a yaw reference $\psi_{\text{ref}}(t)$ which aligns the vehicle heading with the next waypoint. Denote the waypoint sequence by

$$p_i = \begin{bmatrix} p_{1,i} \\ p_{2,i} \\ p_{3,i} \end{bmatrix}, \quad \psi_i, \quad i = 1, \ldots, N, \tag{2}$$

where $N = 16$ is the total number of waypoints in the second-order Hilbert trajectory. Use the waypoint sequence given in Table 1. The horizontal coordinates define a second-order Hilbert curve with branch length 2 m, while the vertical coordinate introduces the required climb-and-descent profile. The yaw waypoints align the vehicle heading with the next position waypoint.

Table 1: Waypoint sequence for the modified second-order Hilbert trajectory.

Waypoint i	$p_{1,i}$ [m]	$p_{2,i}$ [m]	$p_{3,i}$ [m]	ψ_i [deg]
1	0	0	0.00	90
2	0	2	-0.25	90
3	2	2	-0.50	0
4	2	0	-0.75	-90
5	4	0	-1.00	0
6	6	0	-1.00	0
7	6	2	-1.00	90
8	4	2	-1.00	180
9	4	4	-1.00	90
10	6	4	-0.75	0
11	6	6	-0.50	90
12	4	6	-0.25	180
13	2	6	0.00	180
14	2	4	0.00	270
15	0	4	0.00	180
16	0	6	0.00	90

Plot the waypoint sequence in the (p_1, p_2) plane and clearly indicate the order in which the waypoints are traversed. In addition, provide a 3D plot of the waypoint sequence to illustrate the combined horizontal Hilbert pattern and vertical motion.

Answer:

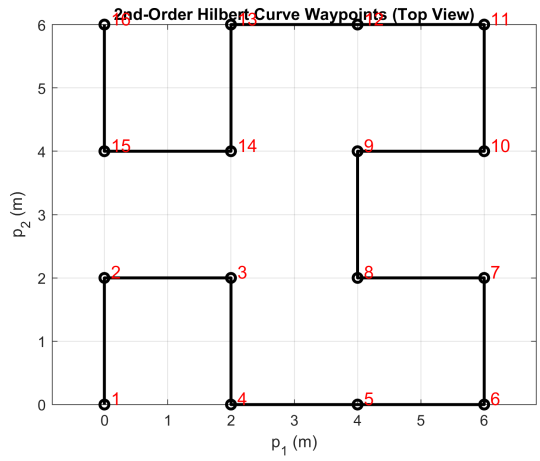

Figure 1: Waypoint sequence in the p_1, p_2 plane in the order of waypoints traversed. The top view visualizes the horizontal Hilbert pattern.

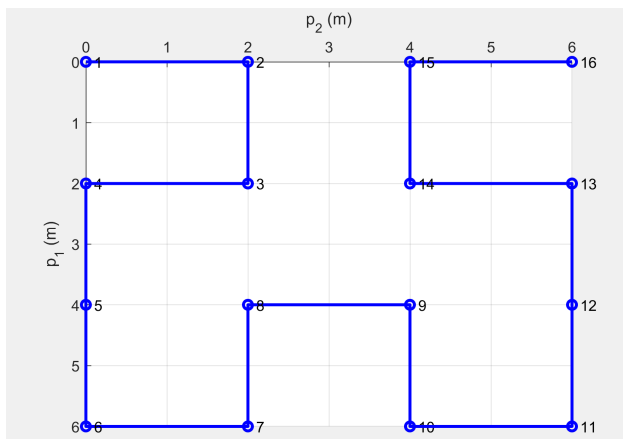

Figure 2: 3D plot of the waypoint sequence in the p_1, p_2 plane in the order of waypoints traversed. The view flipped the axis directions of p_1 and p_2 respectively, to show how the trajectory tracking matches the Hilbert Pattern, as referenced in Figure 3.

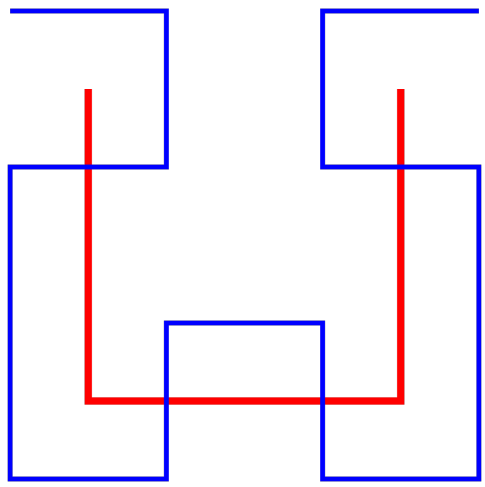

Figure 3: Hilbert Curve. First and Second Orders

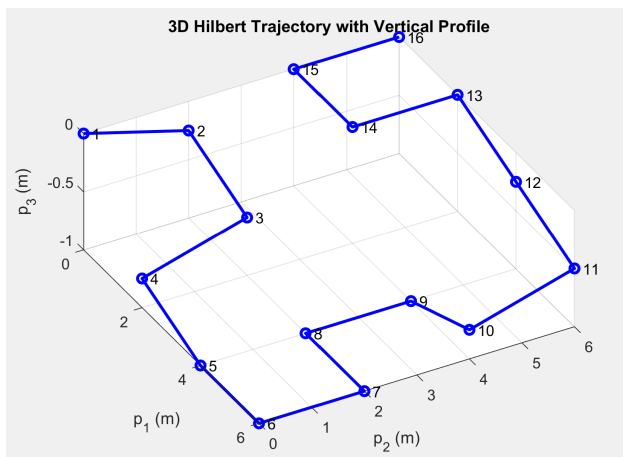

Figure 4: Vertical view of the 3D Hilbert Curve that shows the horizontal and climb-and-descent profile of the vehicle following the waypoints indicated by the table.

2. **Guidance law with velocity and acceleration constraints.** Generate a smooth reference trajectory between successive waypoints using a guidance law with bounded velocity and bounded acceleration as shown in Appendix A of [1] (also seen in class). Set the maximum velocity constraint $v_{p,max} = 1$ m/s and the maximum acceleration constraint $a_{p,max} = 1$ m/s^2. Furthermore, set maximum yaw rate constraint $v_{\psi,max} = 2$ rad/s and the maximum yaw acceleration constraint $a_{\psi,max} = 0.5$ rad/s^2. Consider that the waypoints are reached when the multicopter enters a radius of $R_p = 0.15$ m around the current waypoint. This guidance law produces a trajectory consisting of acceleration, cruise, and deceleration phases.

   Answer: Adding the velocity and acceleration bounds to the model, we can simulate the reference trajectory in 3D with a quiver plot:

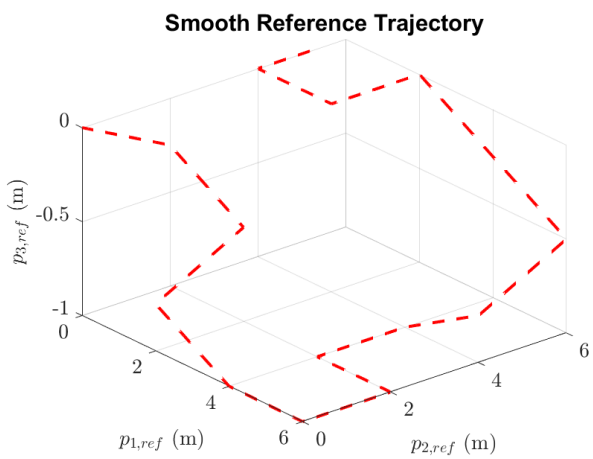

Figure 5: Smooth Reference Trajectory for the 3D Hilbert Curve

3. **Closed-loop simulation of the full nonlinear multicopter dynamics.** Use the cascaded controller developed in the previous steps to track the Hilbert trajectory. In particular, simulate the full nonlinear multicopter dynamics from the previous project, where the outer loop generates the virtual force command, the virtual force is mapped to thrust magnitude and attitude references, and the inner loop generates the torque command. Use the guidance law from the previous part to generate the reference position, velocity, and acceleration along the Hilbert trajectory.

   Answer:

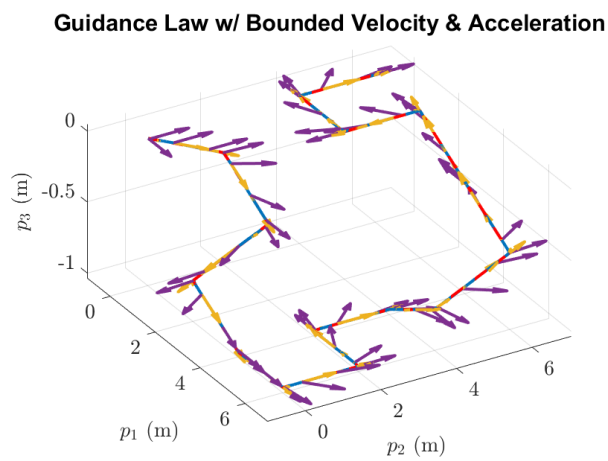

Figure 6: Closed-Loop Simulation With Position and Velocity Vectors

4. **Required plots.**

   (a) The reference and actual trajectories in the horizontal plane (2D trajectory).
   A: Shown below in Figures 7 and a close up in 8.

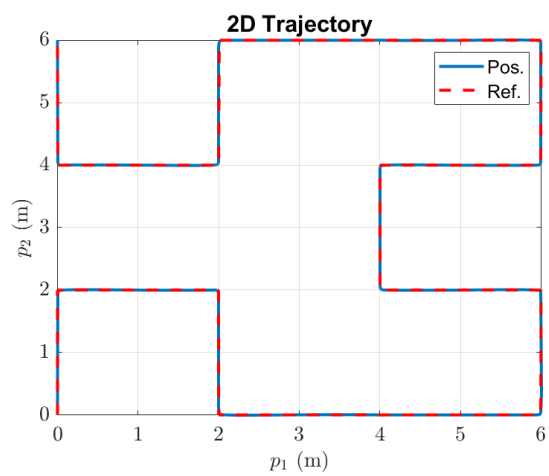

Figure 7: 2D Actual and Reference Trajectories

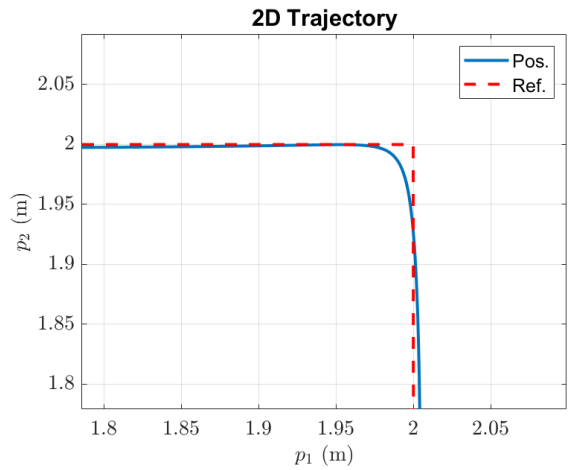

Figure 8: Close up of 2D Actual & Reference Trajectories to Highlight Differences

(b) A 3D trajectory plot using `plot3`. Use `quiver3` to plot the velocity vectors and the yaw, as shown in class.

A: Shown in Figure 6.

(c) The position components $p_1(t)$, $p_2(t)$, $p_3(t)$ with respect to t together with their reference signals (3 subplots).

A: Shown below in Figures 9, 10, and 11 respectively.

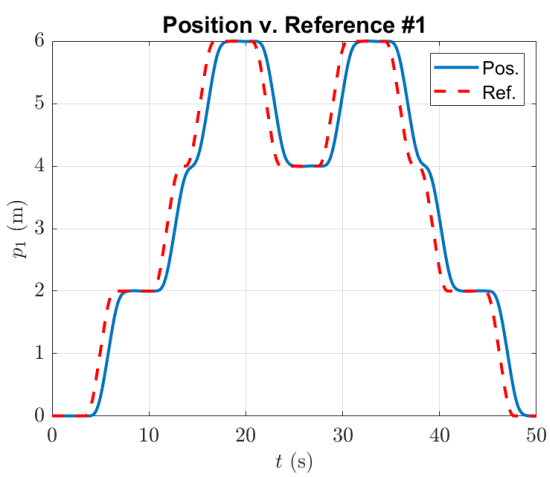

Figure 9: Position vs. Reference for p_1

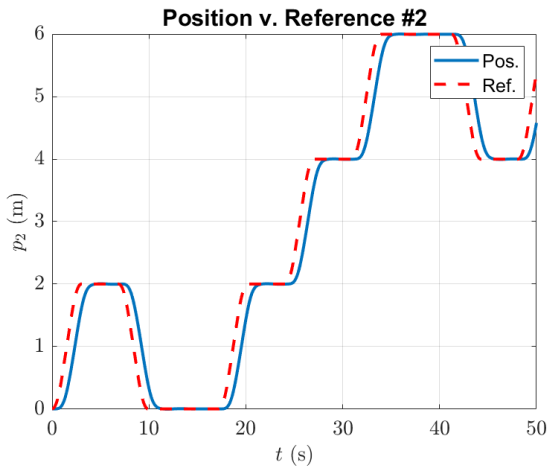

Figure 10: Position vs. Reference for p_2

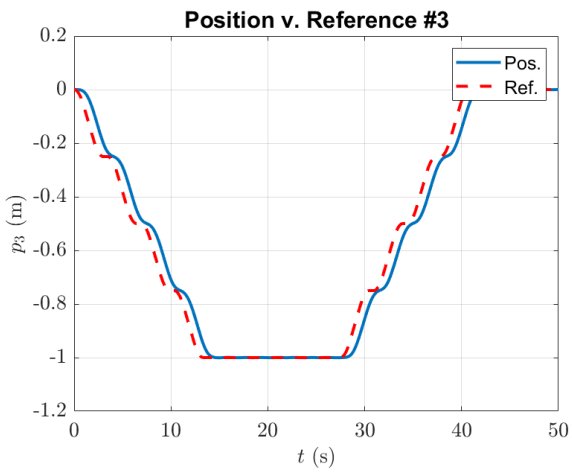

Figure 11: Position vs. Reference for p_3

(d) The tracking errors

$$e_p(t) \triangleq p(t) - p_{ref}(t)$$

A: Shown below in Figure 12.

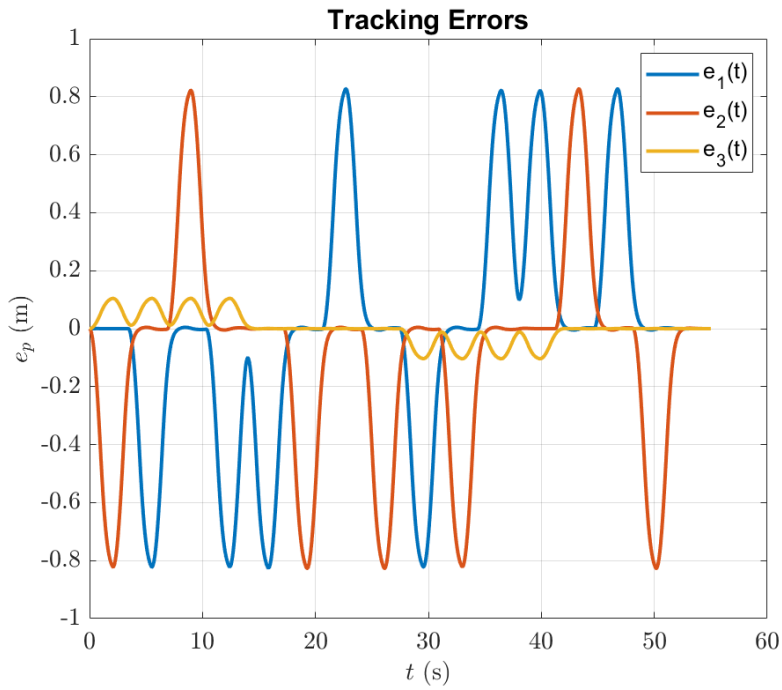

Figure 12: Tracking Errors for p_1, p_2, p_3

(e) The roll and pitch angles together with their commanded reference angles, that is, $\phi(t)$ and $\phi_{ref}(t)$ on one plot, and $\theta(t)$ and $\theta_{ref}(t)$ on another plot.

A: Shown below in Figures 13 and 14.

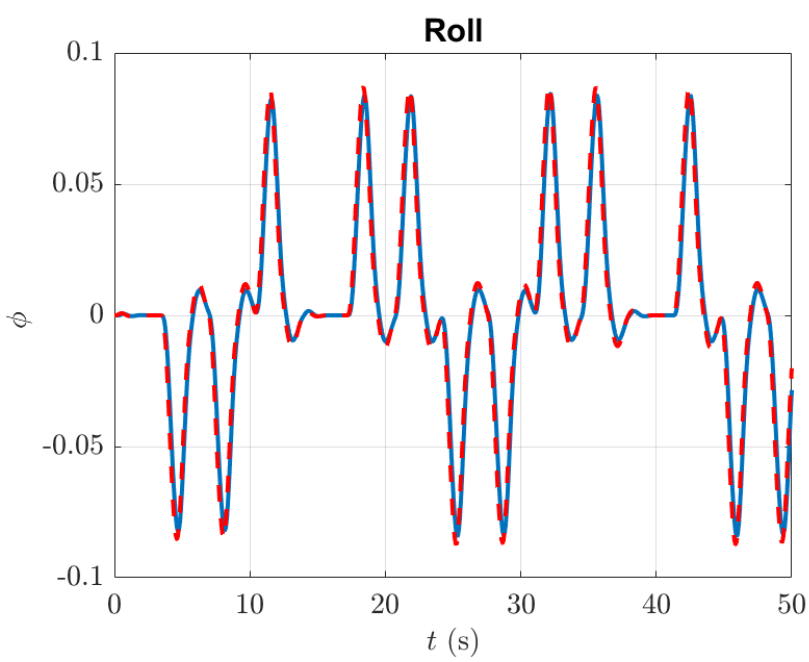

Figure 13: Actual vs. Reference ϕ

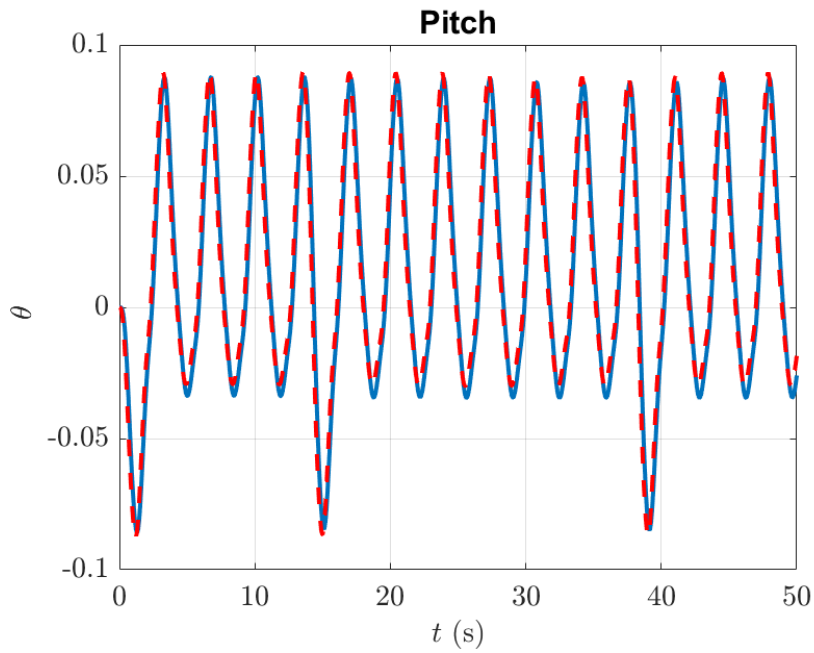

Figure 14: Actual vs. Reference θ

5. **Tracking performance and discussion.** Briefly discuss the closed-loop tracking performance.

   Answer: The largest errors in the tracking happen when the system is accelerating or changing direction quickly. This can be seen in comparing Figures 12 and the position versus reference figures 9, 10, and 11. The effect of sharp turns increases this error, but it is quickly resolved by the system. The inner loop controller tracks the pitch and roll angles sufficiently; however, there is still some delay between the reference trajectory and the actual trajectory from the cascaded design.

Credit Author Statement

Ethan Hardesty: Visualization, Software, Methodology

Christopher Appiah: Coding, Writing, Validation